SMART INDIA HACKATHON 2025

TITLE PAGE

- ***Problem Statement ID- 25015***
- ***Problem Statement - Intelligent Pesticide Sprinkling System Determine by the Infection Level of a Plant***
- ***Theme - Agriculture, FoodTech & Rural Development***
- ***PSCategory- Hardware***
- ***TeamID- 54521***
- ***TeamName(Registeredonportal) - Hardware Hustlers***

AgriLens Kit -A hardware kit for automated spraying of pesticides using AI and smart learning

HARDWARE HUSTLERS

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SIH

AgriLens Kit is a multifunctional, lightweight and farmer friendly solution focused on detecting crop health, diagnosing diseases, and controlling pesticide spraying through an accessible and scalable approach.

Image capture + video capture + sound capture → Preprocessing → K-cluster segmentation (sound and rare diseases) → Stage 1 (crop detection) → Stage 2 (disease classification) → Decision logic & spraying

- **Image Capturing and preprocessing:** Mounted cameras (ESP32-CAM / Smartphone) capture periodic or triggered images of plants. Preprocessing includes RGB→HSV conversion, leaf isolation through green masking and resizing for efficient analysis. A light weight CNN model classifies the captured image into its respective crop(e.g. tomato, potato, rice, wheat).This ensures that the right disease model is selected for the next stage.

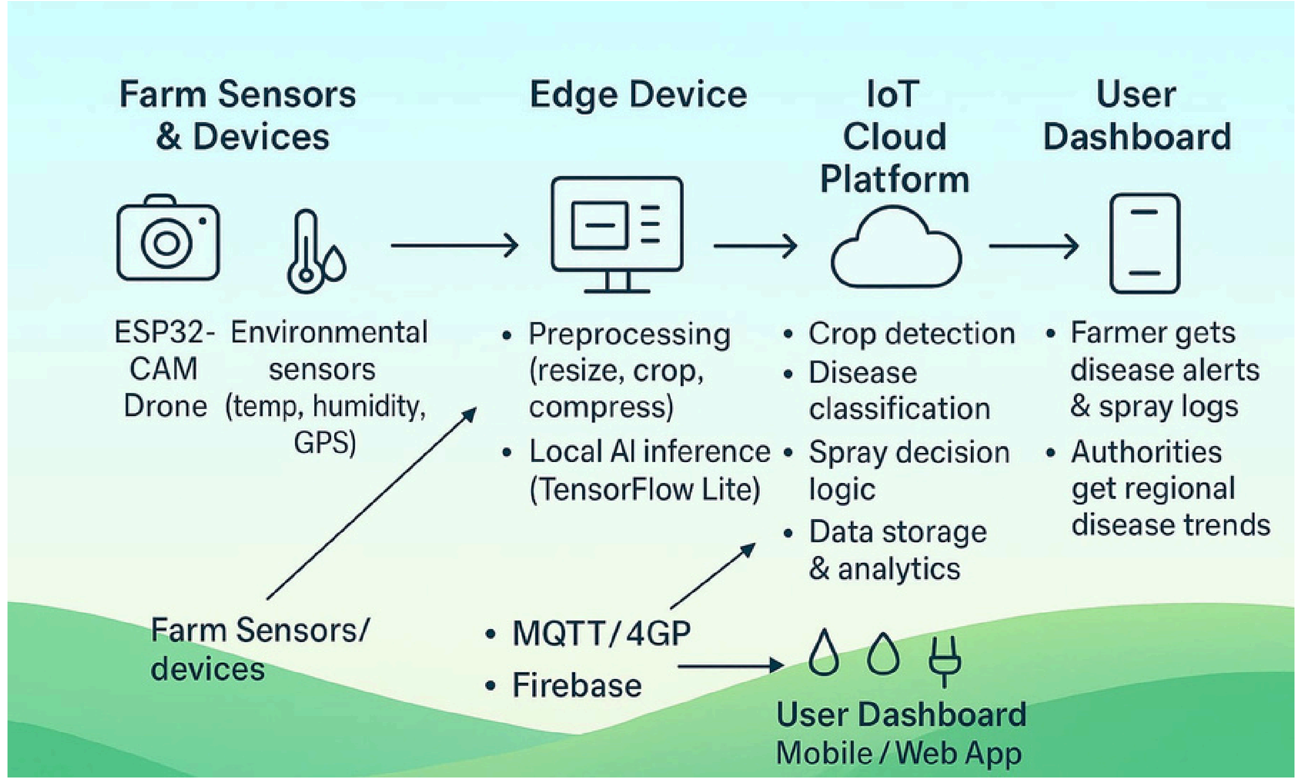
- **Disease diagnosis and Targeted Spraying:** Once the crop is identified, our specialized model analyzes the leaf texture, spots, color and pattern to detect specific diseases(e.g. late blight in potato, leaf curl in tomato, blast in rice). Also soil and water sensors are used to detect if soil is over saturated with pesticides and fertilizers to prevent over spraying. Output includes disease type, severity score, and confidence level.

- **Two Stage model:** Reduces misclassification across crops having the same diseases (eg: blight in potatoes and tomatoes).Easy to add a new crop by plugging in its own classifier. Also provides an unique way to classify the type of disease.

- **Cost Effective and Easy to Use:** The two stage design (crop detection + disease diagnosis) ensures low hardware requirements, scalable deployment and reduced pesticide usage. Also providing the farmers with a mobile application for the mto easily access and control the system.

Plant diseases are classified following FAO-IPPC, EPPO, and CABI Plant wise standards, and severity is measured using a 0-9 disease index scale. This ensures our AI driven detection aligns with international agricultural practices and can be scaled globally, along with insect sound and image processing.

Visual Symptom Mapping: Match detected leaf symptoms (spots, discoloration, blight, mildew, mosaic patterns) with standard descriptions in CABI / EPPO / ICAR guides. Severity Index (Scale 0–5 or 0–9): Using a standard disease severity scale (widely used in FAO and ICAR trials)



TECHNICAL APPROACH

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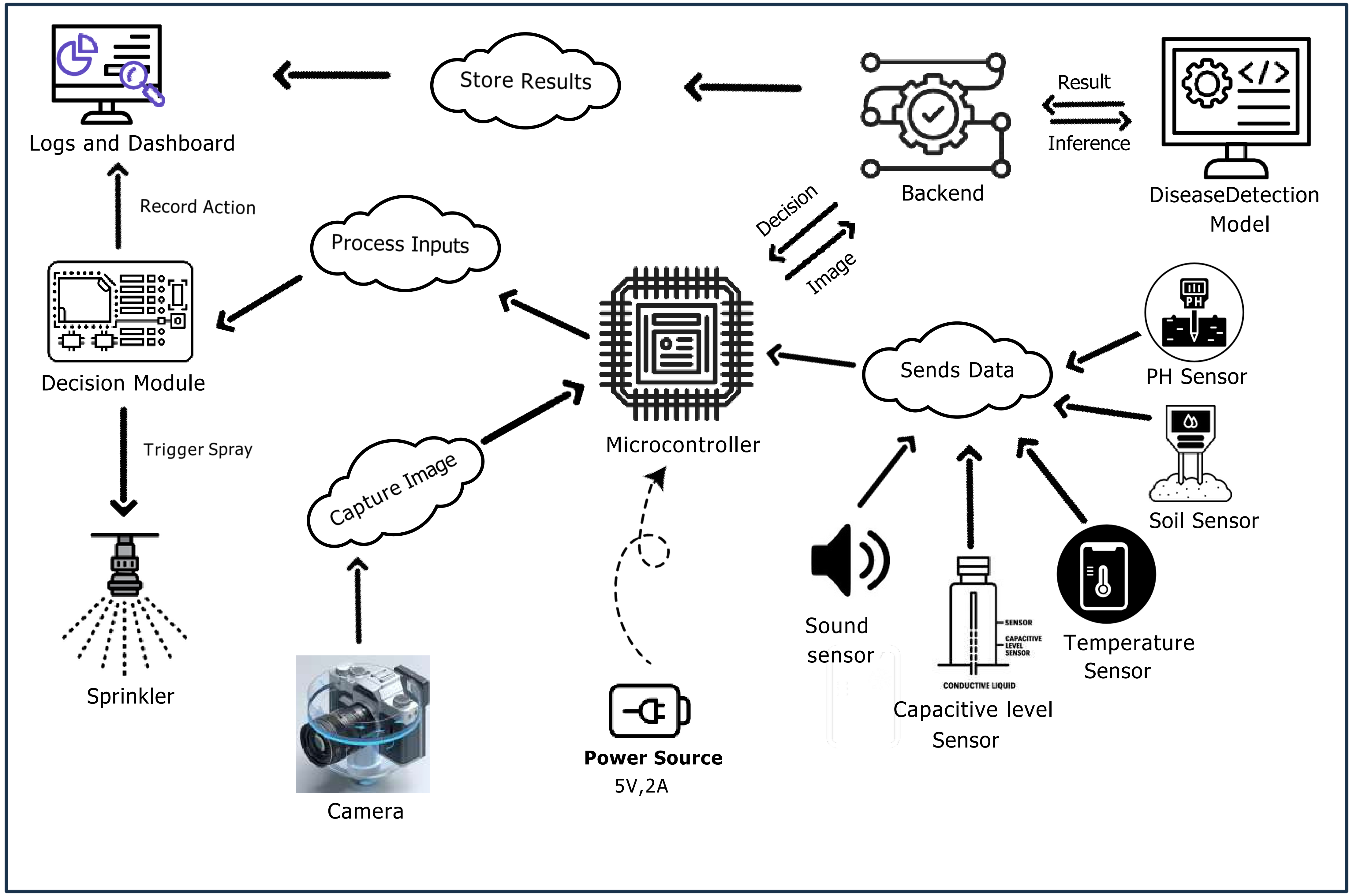
•Process of Implementation:



•Technologies Used:

Technologies used in Hardware:

ThingsBoard, Arduino IDE, LoRaWAN, C++, scada

Tech Stack used in model:

AWS Lambda, PyTorch, TensorFlow, pandas, Python, Flask

Web Application Tech Stack:

Next.js, Express, Tailwind CSS

- Animation of Model (YouTube)
- Datasets of Diseases (Google Drive)
- Prototype in Action (YouTube)
- Explore our Sprinkler model (GitHub)
- Video of Precision Plant care (YouTube)

FEASIBILITY AND VIABILITY

Pictures of prototype:

Viability and Scalability

- **High Market Potential –** Agriculture AI is a rapidly growing sector (projected $4B+ by 2026), with strong demand from farmers, agri-tech startups, and government initiatives.
- **Sustainable & Aligned with Goals –** Supports UN SDG-2 (Zero Hunger), FAO plant health programs, and India's Digital Agriculture Mission 2021–2025, while reducing excessive pesticide use.
- **Integration with Ecosystems –** Can be linked with PMFBY crop insurance schemes, existing farm equipment, and nationwide disease databases for smarter agricultural management.
- **Crop & Geographic Expansion –** Initially focused on potato, tomato, rice, and wheat, but easily extendable to maize, sugarcane, cotton, and horticulture crops; globally scalable once aligned with FAO/EPPO standards.
- **Technology-Driven Growth –** Cloud-based backend, drone & satellite integration, API-based architecture, and continuous farmer feedback ensure adaptability, multi-language support, and smarter AI over time.

IMPACT AND BENEFITS

Impact on target audience:

Accuracy:
The user friendly design, digital outputs, ensure accurate detection of diseases at an early stage preventing large-scale crop damage .

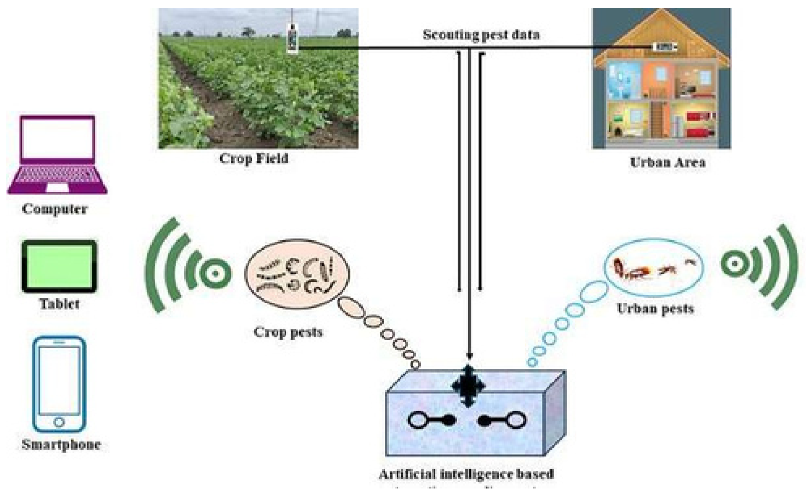
Hardware impact:
IoT connectivity enables real time monitoring and data transfer, making the solution scalable from small farms to large agricultural fields.

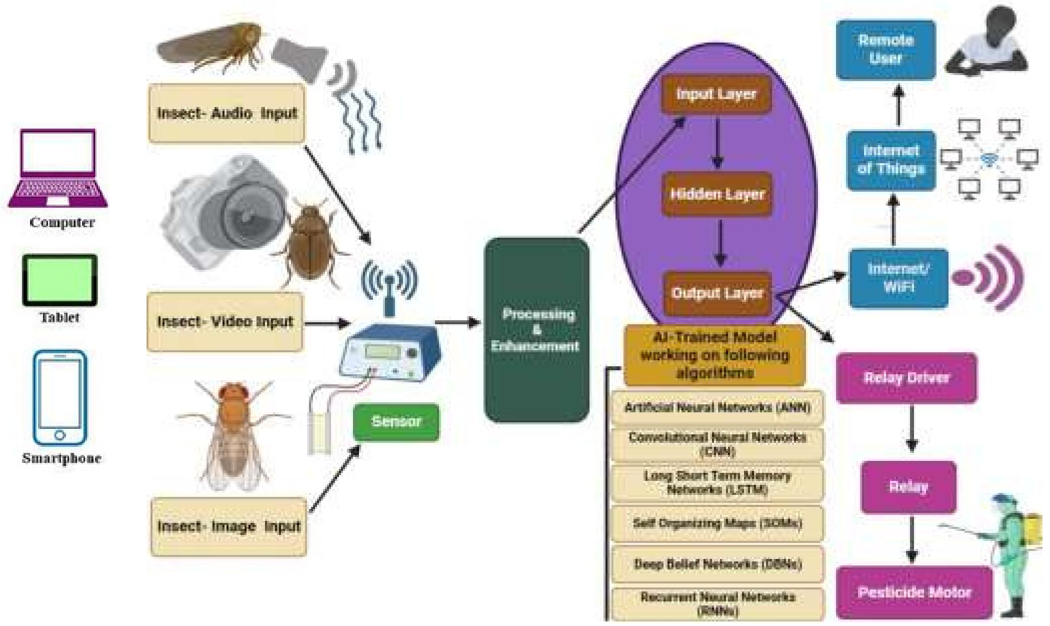
Precision Agriculture:
Sprays pesticides only on infected parts of the plants, which reduces wastage of chemicals and environmental pollution.

Time Saving Automation: Reduce farmer’s manual effort of constant monitoring and spraying. Farmers can manage larger farms with less workforce.

Benefits:

- **Positive effects on health:**
  Minimize the contact with pesticides. AgriLens Kit aims to lower the amount of farmer’s exposure to pesticides in the surrounding environment.
- **Sustainability:**
  Unlike generic systems, it provides crop-specific diagnosis using a two stage AI pipeline, ensuring farmers get accurate and timely insights for potato, tomato, rice, wheat and more.





RESEARCH AND REFERENCES

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2.Disease detection in different plants to save the environment using IOT, image processing and machine learning: a review

3.Mahlein, A.-K. (2016). Plant disease detection by imaging sensors–parallels and specific demands for precision agriculture andplantphenotyping. PlantDisease, 100(2),241–251.

4.Feature engineering to identify plant diseases using image processing —SM Javidan et al., 2024 (ScienceDirect) 5.A

review on automated plant disease detection: motivation, limitations, challenges, and recent advancements for future research —SU Khan, et al., 2025 (Journal of King Saud University)

6.Bock, C. H., & Nutter, F. W. (2011). Detection and measurement of plant disease symptoms using visible-wavelength photography and image analysis.Plant Sciences Reviews,2012, 73.

7. Inspiration: Leaf Disease Detection Using Raspberry Pi(video)